

RAMCO INSTITUTE OF TECHNOLOGY

Department of Mechanical Engineering

Academic Year 2019 – 2020 (Odd Semester)

Innovative practice(s) description

Course code and title : ME6702 Mechatronics

Name of the Instructor(s): Mr.T.Selva Sundar, AP//Mechanical

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Date : 29.07.2019

Activity based learning- 8085 Microprocessor Instruction Set



In the Unit-II (8085 Microprocessor and 8051 Microcontroller) of the subject ME6702 Mechatronics syllabus there is a separate portion which explains the architecture, pin diagram, addressing mode and instruction sets of 8085 Microprocessor. Students are explained these concepts using the 8085 Microprocessor kit available in the Mechatronics Laboratory. For better understanding and easy remembering of the instructions and their functions students are involved in a class room based activity in which students will be able to remember, recall the function of the given instruction of 8085 Microprocessor.

OUTCOME:

1. The students will be able to describe the functions of the Microprocessor 8085 and their interface
2. The students will be able to remember the instruction set of 8085 Microprocessor easily

Sample Activity based learning sheet

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Department of Mechanical Engineering

ME6702 Mechatronics

VII Semester B Section (2019 – 2020)

8085 Instruction set

Instruction	Identify the Category and operation
1. LXI SP, 8 bit data	Data Transfer Instruction $SP \leftarrow 8 \text{ bit data}$ ✓
2. MVI R, FFH	Data Transfer Instruction $R \leftarrow FFH$ ✓
3. ADC M	Arithmetic Instruction $A \leftarrow A + M + CM$ ✓
4. INR rp	Arithmetic Instruction $rp \leftarrow rp + 1$ ✓ $rp \leftarrow rp - 1$ if no flag affected
5. CMP R	Logical Instruction $A - R$ ✓
6. RLC	Logical Instruction r-take left without carry ✓
7. STC	Logical Instruction Set carry $cy \leftarrow 1$ ✓
8. JPE	Branching Instruction Jump if $P = 1$ ✓
9. CM	Branching Instruction call memory $CMR - R; CMR$ ✓
10. RZ	Branching Instruction Return ? ✓
11. IN 8 bit data	Initiate input operator ✓ type of instruction I/O
12. EI	Enable Interrupt [All are enabled] Machine related instruction ✓

G. Pooragavan
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Name of the students

4/1 29/11/19
Signature of the Faculty In charge